



A Survey of Project Selection in Tourist Spots Using Smart City Platforms

Surja Sanyal^{1,2}(✉) , Sajal Mukhopadhyay³ , Fatos Xhafa⁴ ,
and Jaya Mukhopadhyay⁵

¹ Indian Institute of Technology Guwahati, Guwahati 781039, Assam, India
surja.sanyal@iitg.ac.in, surja.sanyal@iitbhu.ac.in

² Indian Institute of Technology (BHU) Varanasi, Varanasi 221005, Uttar Pradesh, India

³ National Institute of Technology Durgapur, Durgapur 713209, West Bengal, India
smukhopadhyay.cse@nitdgp.ac.in

⁴ Universitat Politècnica de Catalunya, Barcelona 08034, Catalonia, Spain
fatos@cs.upc.edu

⁵ Academy of Technology, Hooghly 712121, West Bengal, India
jaya.mukhopadhyay@aot.edu.in

Abstract. Tourism is a valuable financial support for Governments worldwide. Therefore, Governments promote it by undertaking new facility construction projects for tourist spots. These projects are to be selected from a project pool that satisfies a budget constraint. To make this selection process transparent, the Government often involves the public using technological platforms available in existing and upcoming smart cities. Several mechanisms have been proposed to date to either optimize or extend the process's functionality. This article discusses the project selection process in the smart city context, then formalizes the problem, and provides a detailed analysis and performance evaluation of recent works that either optimize or extend its functionality. This article concludes by summarizing the challenges, including over-tourism, open issues, and directions for future work in this domain.

Keywords: Tourism · Administration · Project Selection · Smart Cities · Crowdsourcing · Decision Making · Budget Feasibility · Preference Aggregation · Weighted Voting · Frequency Awareness · Explainable AI · Tourist Incentive

1 Introduction

Several countries rely on tourism as their main source of income [1]. The Governments of these countries intend to support and promote this industry by providing new facilities in the tourist spots. These facility creation projects are selected from a pool of potential projects for the tourist spot under consideration. Traditionally, the Government involves decision-makers internally (hereafter, ministers) in selecting the projects to undertake. However, sustainable

tourism development requires the residents of the tourist spot to be supportive and contribute to the cause [2,3]. Furthermore, tourists visiting the site also provide valuable feedback and suggestions to improve their experience [3]. Therefore, the Government has recently been involving them in the process as well to promote transparency, understand public sentiment, and gather useful insights and experiences [4]. To this end, Artificial Intelligence (AI) and crowdsourcing have provided a promising platforms for the involvement of residents and tourists in the decision-making process as a financially friendly mode for monitoring and management of remote systems [5–9]. Such technologies are already present in existing and upcoming smart cities and help promote the mass percolation of the process and its awareness, benefiting the tourism industry directly and indirectly [9–12]. Decision-making for the tourism industry using AI-based mechanisms and for the aggregation of AI-generated preferences using social choice-based algorithms have been well researched [13–16].

Although society benefits in various ways from the tourism industry, a substantial protest against over-tourism is currently underway in several European nations [17]. Residents of overcrowded tourist spots have pointed out that the industry is putting pressure on the climate, infrastructure, and daily life [17,18]. Involving residents and tourists in industry decision-making processes provides the administration with valuable insights in resolving potential issues by majority preference aggregation [2]. Additionally, it allows residents and tourists to voice their views on whether they want to participate in protests against over-tourism [19]. The project selection process involves tourists, residents, and ministers. Each stakeholder provides their preferences for the projects in the project pool. The aim of the project selection process is to generate a list of projects from this pool to be undertaken by the Government, ensuring the budget constraint is met, and stakeholders' preferences are taken into account. Surprisingly, very few works have been done to optimize this process or to extend its functionality beyond the basics mentioned above. Additionally, surveys on the project selection process are negligible in number.

Therefore, this article aims to provide a comprehensive survey of the latest works on optimizing the project selection process and extending its functionality. The works discussed in this article utilize technologies available in existing and upcoming smart cities, such as crowdsourcing and Explainable AI-based tools, to fine-tune the project selection process. Additionally, this article evaluates the performance of the latest state-of-the-art mechanisms in this research area. Furthermore, this article presents the future research scopes, challenges, and open issues of the project selection process. In summary, the major contributions of this article are as follows:

- At the beginning, a step-by-step overview of the project selection process has been provided, highlighting its challenges and issues.
- A detailed literature review and state-of-the-art survey of the latest works related to the project selection process in smart cities has been provided.
- The performance of a few recently published works related to the optimization and functionality extension of the project selection process is analyzed.

- Finally, the future research scopes, challenges, and open issues of the project selection process in the context of smart cities are discussed.

The remainder of this article is organized as follows. Section 2 presents the overview of the project selection process. The latest works in stakeholder preference aggregation are discussed in Sect. 3. The performance of the discussed mechanisms is analyzed in Sect. 4, followed by a discussion of future scope in Sect. 5. The survey is summarized, and a conclusion is drawn in Sect. 6.

2 Project Selection Process Overview

The project selection process for tourist spots involves three stakeholder groups. Firstly, the traditional stakeholders are the ministers who are internal to the Government body. They are the fewest in number when compared to other stakeholder cardinalities. Secondly, the residents of the land form the majority of the total stakeholders in terms of numbers. Lastly, tourists are the most experienced stakeholders regarding the gaps in the current tourism infrastructure. Additionally, the process involves a public list of potential projects that the Government can undertake in the tourist spot under consideration. Each of these projects has an associated cost. The purpose of the process is to produce a list of projects from the project pool for the Government to undertake. However, the total cost of all Government projects must not exceed the available budget. The preferences ν of the stakeholders $s \in S$ are submitted to the system at the onset. The preferences are ranked lists over the available projects P for the tourist spot under discussion. It is assumed that the stakeholders provide their lists in a strictly decreasing order of preference. That is, no two projects have the same preference, or ranks, for the same stakeholder $s \in S$.

$$\therefore p_i \neq p_j \implies \nu_s^{p_i} \neq \nu_s^{p_j}; \forall s \in S, \forall p_i, p_j \in P. \quad (1)$$

The list V^P of all preference positions for each project $p \in P$ is the input to the aggregation mechanism that produces a single list V^P for all available projects P ordered in decreasing order of preference positions (or ranks) obtained.

$$V^P = \sum_{p=1}^P V^p = \sum_{p=1}^P \sum_{s=1}^S \nu_s^p. \quad (2)$$

The selected list of projects $Q = \{q_1, q_2, \dots, q_k\}$ for the Government to undertake is then populated by selecting the top-ranked k projects that exhaust the budget, as in a knapsack problem. Ties between projects are resolved arbitrarily during project selection, and partial projects can be considered as the last project. It is worth noting that stakeholders' preferences may take many forms, not just a ranked list. For example, traditional single-ranked preferences can be used as an input structure. Single-peaked preferences are also a viable choice for preference submission and have been considered the preferred input structure in the literature.

Mass percolation and extensive participation in the preference submission process heavily depend on the technological platforms available in existing and upcoming smart cities [10, 11]. AI-driven decision-making and AI-generated response aggregation are well-studied use cases for such preference-aggregation processes [9, 16]. The Internet of Things platform has been shown to be pivotal for task crowdsourcing [6, 12]. Crowdsourced data uses participants' mobile devices for various applications [20]. Resource-intensive tasks, such as data collection and labeling across diverse use cases, have been successfully crowdsourced to participants [8]. Various open challenges have also been addressed through crowdsourcing, including decision-making use cases such as project selection in tourist spots [21, 22].

3 Aggregation of Stakeholder Preferences

The literature has been obtained from the *Google Scholar* database using the keywords *tourist spot selection*, *weighted voting*, *preference aggregation*, and *decision making*. On several occasions, these searches produced overlapping sets of articles. However, this domain of research remains underexplored, yielding only a few relevant articles. The latest and most relevant works in this domain are presented next, along with their strengths and weaknesses.

3.1 The RFE–MV Mechanism

RFE–MV [16] solves an AI-based decision-making problem for attack detection using majority voting as an aggregation method. The stakeholders for this work are AI-based classifier networks. These networks provide a single output class per stakeholder. These outputs are accepted as input to the aggregation mechanism, and a single class is produced as its output. Under the hood, the aggregation mechanism uses a majority rule-based voting method. The class with the highest input count is selected as the aggregated class. However, the input classes are not equally likely to be submitted by stakeholders, as observed in real-life scenarios. Some classes in the input class range are more probable to be submitted to the aggregator than others. Although the majority vote aggregation mechanism is used for attack classification with two classes, it generalizes well to multiclass preference submissions, where a list of aggregated classes is expected as output. However, this mechanism does not account for the different stakeholder groups observed in real-life scenarios and treats all stakeholders equally, which does not closely resemble real-life scenarios.

3.2 The TODM–AVR and TODM–MVR Mechanisms

TODM–AVR and TODM–MVR [15] rely on the average and the median voting rules, respectively. The stakeholders for these mechanisms are the residents of the place. The Government involves the residents in the decision-making process through crowdsourcing. The aim of the process is to select a spot for facility construction. The eligible locations for facility construction are mapped as a straight line, and each resident's preference is represented by a single peak along this line. The aggregation of these single-peaked preferences has been performed using two approaches. Firstly, the aggregation has been performed using the average voting rule (TODM–AVR), where the average of all preference peaks is reported as the selected location for facility construction. However, this mechanism is shown to be easily deceived by stakeholders who misreport their preferences to gain on the final outcome. Therefore, the second approach for aggregation has been the median voting rule (TODM–MVR). Using this rule, the median of all preference peaks has been reported as the selected location for facility construction. This approach has been shown to resist misreports by stakeholders and has been upheld as the better mechanism of the two discussed herein. Although the median voting rule is used by the aggregation mechanism for single-peaked stakeholder preferences, it generalizes well to ranked-list preference submissions, where a list of aggregated ranks is expected as output. However, this mechanism does not account for the different stakeholder groups observed in real-life scenarios and treats all stakeholders equally, which does not closely resemble real-life scenarios.

3.3 The PSTRM–KV Mechanism

PSTRM–KV [14] presents a crowdsourced preference aggregation mechanism to perform the project selection process. In this mechanism, three stakeholder groups have been considered: tourists, residents, and ministers. Each of these stakeholders provides a preference list over the available projects for the system to aggregate. The aggregation mechanism assigns different weights to the preferences of different stakeholder groups. The tourists and the residents have been assigned the same weight for their preferences. However, the ministers have been assigned a higher weight to their preferences for the available projects. The system then aggregates these preferences using the assigned weights for each preference group to produce a single weighted preference list for the selected projects. This weighted preference list may include one or more preferred projects with the same aggregated weighted preference. Projects are selected for undertaking by the Government from this aggregated weighted project list from the most preferred to the least preferred. This process is modeled as a Knapsack problem, and the projects are selected until the available budget is exhausted. Arbitrary tie-breaking mechanisms were used to determine the order of projects with the same aggregated weighted preference. The last project is either partially funded or not selected for undertaking. It has been shown that PSTRM–KV accounts

for the differences in stakeholder priorities that closely reflect real-life scenarios. Furthermore, PSTRM-KV shows that differences in priorities can alter the outcomes of the preference aggregation and project selection processes. However, PSTRM-KV has considered static weights predetermined by the system administrator and has not considered contribution-based weights for the different stakeholders. This mandates that the administrator tune the weights for the stakeholder groups each time voting takes place.

3.4 The FIREBOLT Mechanism

FIREBOLT [13] presents a crowdsourced preference aggregation mechanism, similar to PSTRM-KV, for the tourist selection process. The aim of this mechanism is to produce an aggregated, weighted preference list from which the Government will choose the projects to undertake, satisfying the budget constraint. However, FIREBOLT also accepts as input the cardinality of each stakeholder group and the frequency of visits by the tourist stakeholder group to generate the aggregated list of projects. The mechanism uses the cardinality of the stakeholder groups to compute an Explainable AI-based Shapley score for each group. This score is then used as the weight for the preference aggregation of that stakeholder group. The FIREBOLT mechanism further uses tourists' visit frequencies to group them internally into equal-sized frequency buckets. Thereafter, Shapley scores are computed for these buckets using the frequency of tourists' visits to each bucket. These bucketed Shapley scores are then used as the weights for the individual buckets, based on their contributions to the voting process. These bucket weights are scaled to account for their cardinality and then multiplied by the weight of the tourist stakeholder group while preparing the overall weighted preference list.

FIREBOLT also introduces contribution-based incentives for tourists by multiplying scaled, bucketed Shapley scores by an incentive factor set by the system administrator. Thus, FIREBOLT addresses the static preference-weight issue and accounts for differences in stakeholder group cardinality and individual tourists' varying visit frequencies using Explainable AI-based Shapley scores. However, it uses buckets while computing frequency-based tourist weights. This approach reduces the system's computational load but introduces weight approximations for tourists with different visit frequencies in the same bucket. This may result in a different outcome than when each individual frequency is used in their weight calculation rather than bucketing them. Additionally, because the Shapley score calculation has exponential time complexity, only a small number of buckets can be used to accommodate the ever-increasing frequency range of tourists visiting the place.

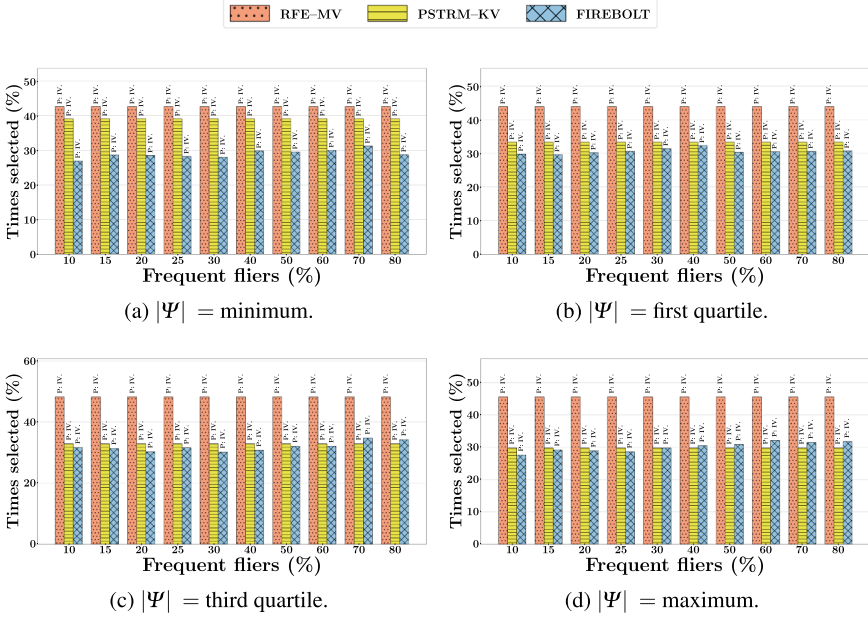


Fig. 1. Percentage of times the most frequently selected first project got selected across all simulations of the mechanisms for different percentages of frequent fliers during honest voting for different numbers of ministers $|\Psi|$ in the variation range.

4 Performance Evaluation

The analysis in this section presents the frequency of selecting projects across different positions in the selected project list, averaged over all experiments, plotted against various values of frequent flier tourists. The results presented in Figs. 1, 2, 3, and 4 show both honest and dishonest voting by stakeholders. The results are presented as percentages of the maximum possible value in each category. The results show variations in the minimum, maximum, first, and third quartiles of stakeholder group cardinality. Selected projects are prefixed with “P:” with a Roman numeral for the project name from the list of available projects. For example, “P: IV” is the fourth project from the list of available projects selected for the Government to undertake.

4.1 Experiment Environment and Setup

Real data, taken from [OPEN DATA BARCELONA](#), was cast to the simulation settings using Python libraries. The total number of ministers ranged from 1 to 10, the total number of tourists ranged from 10 to 100, and the total number of residents ranged from 10 to 100. The project pool had 20 projects with various

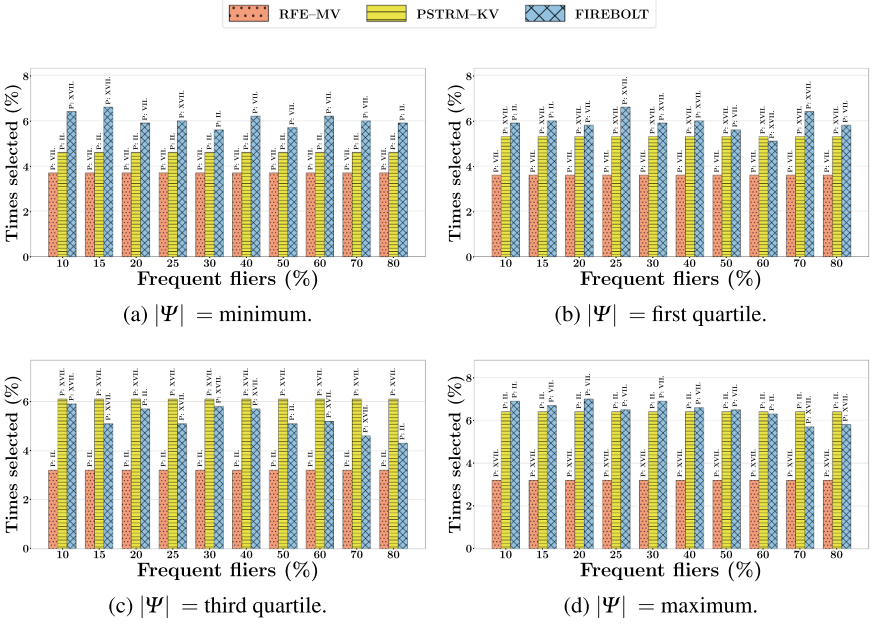


Fig. 2. Percentage of times the most frequently selected fifth project got selected across all simulations of the mechanisms for different percentages of frequent fliers during honest voting for different numbers of ministers $|\Psi|$ in the variation range.

costs. A total budget of \$500000 was made available. A cap of 100 visits was applied to all tourists. Furthermore, frequent visitors were the tourists who visited at least 80% of this cap. Frequent visitors ranged from 10% to 80% of the total group of tourists. The comparison was done against the three most capable state-of-the-art mechanisms among those discussed, namely, FIREBOLT [13], PSTRM-KV [14], and RFE-MV [16]. For PSTRM-KV, the weights for residents' and tourists' votes were set to 1, and those for the ministers were set to 10.

4.2 Experiment Results and Analysis

Figure 1 presents the percentage of times the most frequently selected first project got selected across all simulations of the mechanisms for different percentages of frequent fliers during honest voting for different numbers of ministers $|\Psi|$ in the variation range. It can be observed that all mechanisms selected the same project as their first choice. This is because the mechanisms performed equivalently for heavily voted projects. Figure 2 presents the percentage of times the most frequently selected fifth project got selected across all simulations of the mechanisms for different percentages of frequent fliers during honest voting for different numbers of ministers $|\Psi|$ in the variation range. It can be observed that

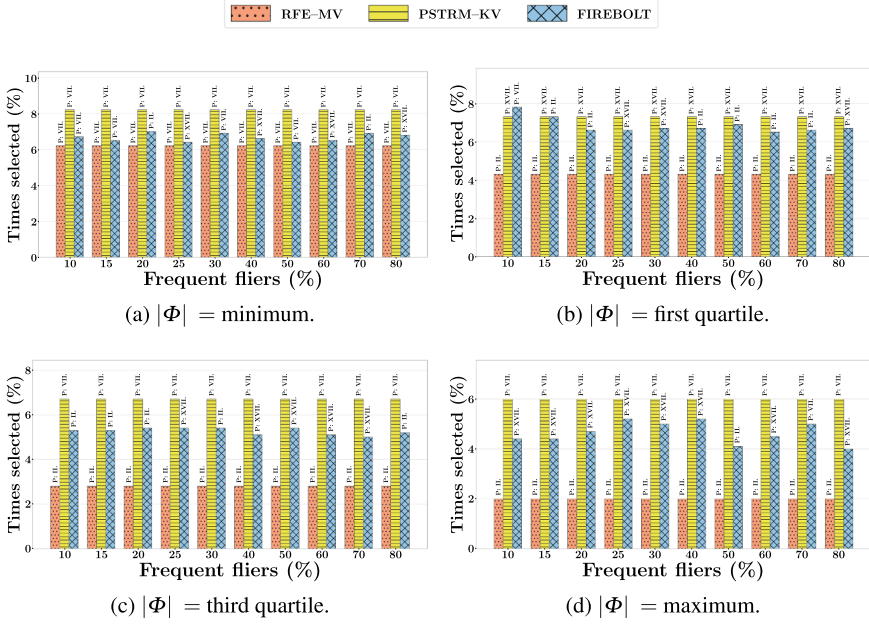


Fig. 3. Percentage of times the most frequently selected fifth project got selected across all simulations of the mechanisms for different percentages of frequent fliers during dishonest voting for different numbers of residents $|\Phi|$ in the variation range.

the diversity in the fifth selected project was higher than that of the first selected project shown in Fig. 1. This is because the differences in the aggregated votes for the fifth project position were lower than those for the first project position, leading to more frequent selection changes and lower selection percentages for individual projects. Additionally, FIREBOLT captured the effect of variation in the percentage of frequent fliers on the diversity of the fifth selected project.

For the dishonest voting scenario, Fig. 3 presents the percentage of times the most frequently selected fifth project was selected across all simulations of the mechanisms, for different percentages of frequent fliers and different numbers of residents $|\Phi|$. It is observed that FIREBOLT captures the effect of variation in the percentage of frequent fliers on diversity in the fifth selected project, as the number of residents varies. Figure 4 presents the percentage of times the most frequently selected fifth project was selected across all simulations of the mechanisms, for different percentages of frequent fliers during dishonest voting, for different numbers of ministers ($|\Psi|$). It can be observed that FIREBOLT captured the effect of variation in the percentage of frequent fliers on diversity in the fifth selected project, when the number of tourists and ministers varies. Thus, FIREBOLT captured the effect of variation in the percentage of frequent fliers more accurately than the other mechanisms in both honest and dishonest voting scenarios, making it a preferred option for the process.

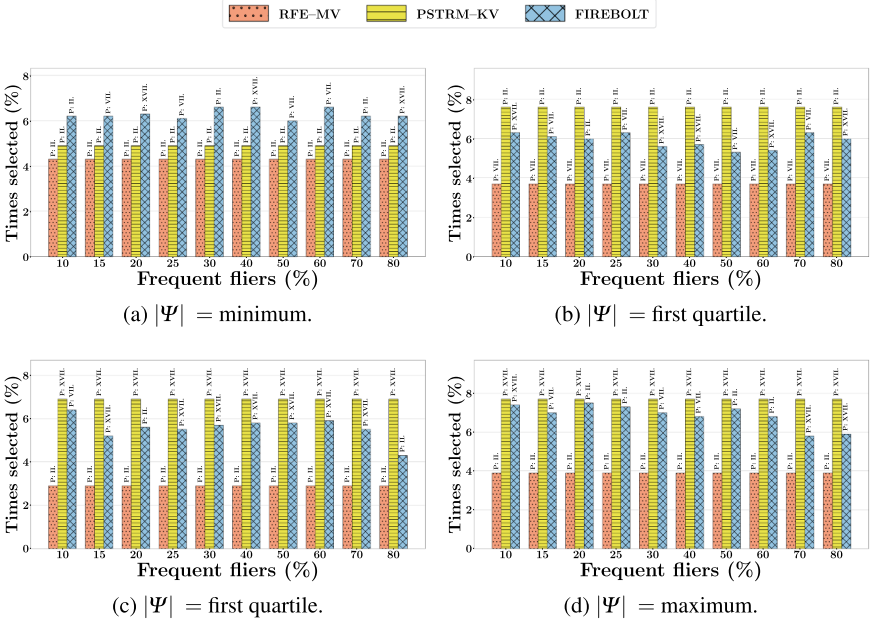


Fig. 4. Percentage of times the most frequently selected fifth project got selected across all simulations of the mechanisms for different percentages of frequent fliers during dishonest voting for different numbers of ministers $|\Psi|$ in the variation range.

5 Challenges and Open Issues

Several challenges and issues remain in this domain of research, as well as its future scope, that require the immediate attention of researchers and policymakers.

- Although society benefits in many ways from the tourism industry, a substantial protest against over-tourism is currently underway in several European nations. There is a need for in-depth research on the negative impacts of over-tourism.
- Additionally, none of the aforementioned works presents AI-based learning tools for predicting the number of tourists participating in the voting process at any given instant. AI-based predictive learning extensions are promising avenues for advancing existing work in the domain and a viable direction for future work.
- Furthermore, computational cost is an open issue that future research must focus on. Since many mechanisms proposed in the literature have algorithmic components with exponential computational complexity, an investigation of their computational cost is needed for the completeness of their performance analysis.

- Moreover, the current literature heavily focuses on achieving optimal outcomes at the expense of computational complexity. An alternate heuristic approach, assisted by AI-based modeling and capable of achieving results close to the optimal while incurring lower costs, is another promising direction for future work.
- Lastly, frequent visitors can be tracked and prompted to participate in voting whenever it takes place. Using present-day smart city technologies, such as AI-based profile and preference monitoring tools, preemptive participation can be encouraged to enhance engagement and improve the tourism experience.

6 Conclusion

In many cases, it is evident that the tourism industry generates revenue for Governments. Therefore, the Government promotes it by undertaking several projects from a pool of available ones under a budget constraint. The Government involves tourists, residents, and ministers in a preference submission process. The aggregation of preferences aims to obtain a ranked list of projects in decreasing order of preference, from which the projects to be undertaken are selected until the available budget is exhausted. This article surveyed existing work on optimizing performance and enhancing functionality of this aggregation process, leveraging the technological platforms available in smart cities. Also, this article provided a performance analysis of the leading state-of-the-art mechanisms in the domain. Additionally, this article examined how over-tourism can have adverse effects on the social lives of people in a specific region of a country. Furthermore, this article provided a brief overview of the existing challenges, open issues, and potential directions for future work in this domain, which require attention from researchers and policymakers.

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